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```python
# Verify GPU availability
print("Num GPUs Available: ", len(tf.config.list_physical_devices('GPU')))
```

```
Num GPUs Available:  1
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[19]:

```python
import os
import glob
import numpy as np
import tensorflow as tf
from tensorflow.keras import layers, models
import matplotlib.pyplot as plt

# ==========================================
# 1. GPU VERIFICATION
# ==========================================
print("--- Verifying GPU Hardware ---")
gpus = tf.config.list_physical_devices('GPU')
print("Num GPUs Available: ", len(gpus))
if len(gpus) > 0:
    print("TensorFlow will train using your GPU.")
else:
    print("WARNING: No GPU detected. Training will defaults to CPU.")
print("-" * 30)

# ==========================================
# 2. DATA PATHS & HYPERPARAMETERS
# ==========================================
# Adjusted specifically to match your nested extraction path
TRAIN_DIR = '/workspace/data/seg_train/seg_train'
TEST_DIR = '/workspace/data/seg_test/seg_test'
PRED_DIR = '/workspace/data/seg_pred/seg_pred'

IMG_SIZE = (150, 150)
BATCH_SIZE = 32
EPOCHS = 20

# ==========================================
# 3. LOADING THE DATASETS
# ==========================================
print("Loading Training Dataset...")
train_dataset = tf.keras.utils.image_dataset_from_directory(
    TRAIN_DIR,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    label_mode='categorical'
```

Mode: Command

0 1 Python 3 (ipykernel) | Idle

Search

DELL